SCOPE 2 · GHG PROTOCOL · MALAYSIA

Scope 2 Emissions Checker

Cleans messy electricity bills, flags what a human must review, and calculates emissions from official Malaysian grid factors.

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Bills arrive messy, and errors flow into GHG reports

Mixed formats

kWh, kwh, MWh and kW. “13,250” as text. Dates written two ways.

Dirty records

Duplicate rows, site-name typos, missing usage, missing months.

Suspicious values

A 10x spike from an extra digit. A negative reading. Power entered as energy.

Python computes. AI drafts. Code checks the AI.

1 · CLEAN

Fix units, dates, names. Every automatic change is logged.

2 · FLAG

Spikes, gaps, negatives, bill mismatches. Flag, never fix.

3 · CALCULATE

kWh × factor ÷ 1,000, on verified rows only.

4 · VERIFY

Claude drafts the summary. A script checks every number.

Measured, not claimed

11/11
planted errors caught, with no false alarms

0
flagged rows leaked into the emission totals

425.5
tCO2e from 19 of 24 site-months (79.2% coverage)

Factor: 0.740 kgCO2e/kWh, Peninsular Malaysia 2024, provisional (Energy Commission). Source printed beside every result.

What broke, and what it taught me

The verifier flagged “150”

I had pasted my own prompt into the summary cell. The check caught a number that was not in the data.

The lesson

I changed the Factor Year from hardcode to live-code.

Honest limits

Data

No 2025 factor published yet, so 2024 is used with a printed warning. Factors are provisional.

Testing

Checked against one dataset I designed, not unseen files. Thresholds (3x, 25%) are judgment calls.

Scope

Location-based Scope 2 only. No Scope 1, Scope 3 or market-based method.

Next: from a notebook to a reusable skill

1. Test on unseen files and track false positives
2. Package the cleaning and checks as a Claude Code skill
3. Pull bills from email or a shared drive, not CSV upload
4. Keep the rule: flag, never guess